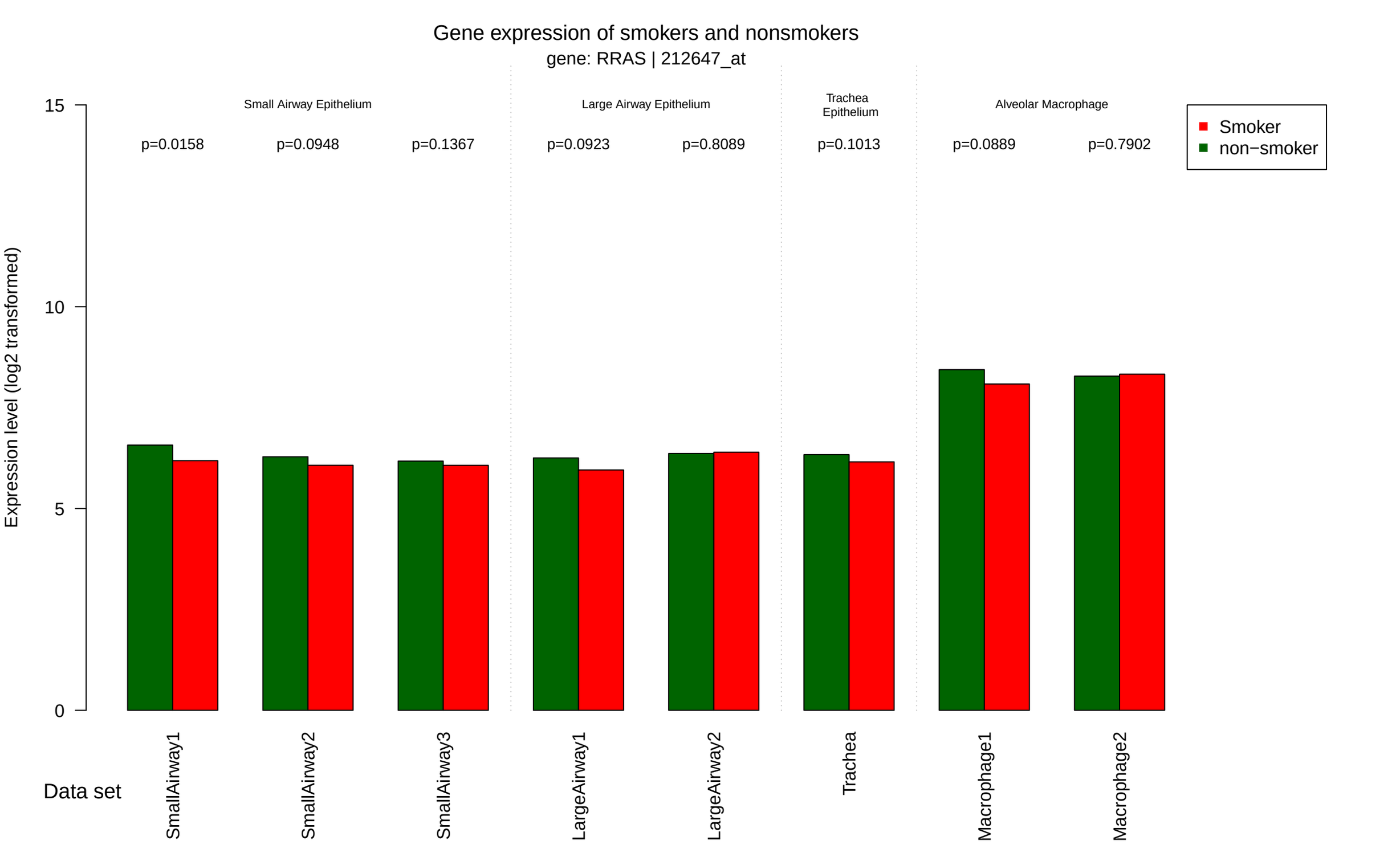
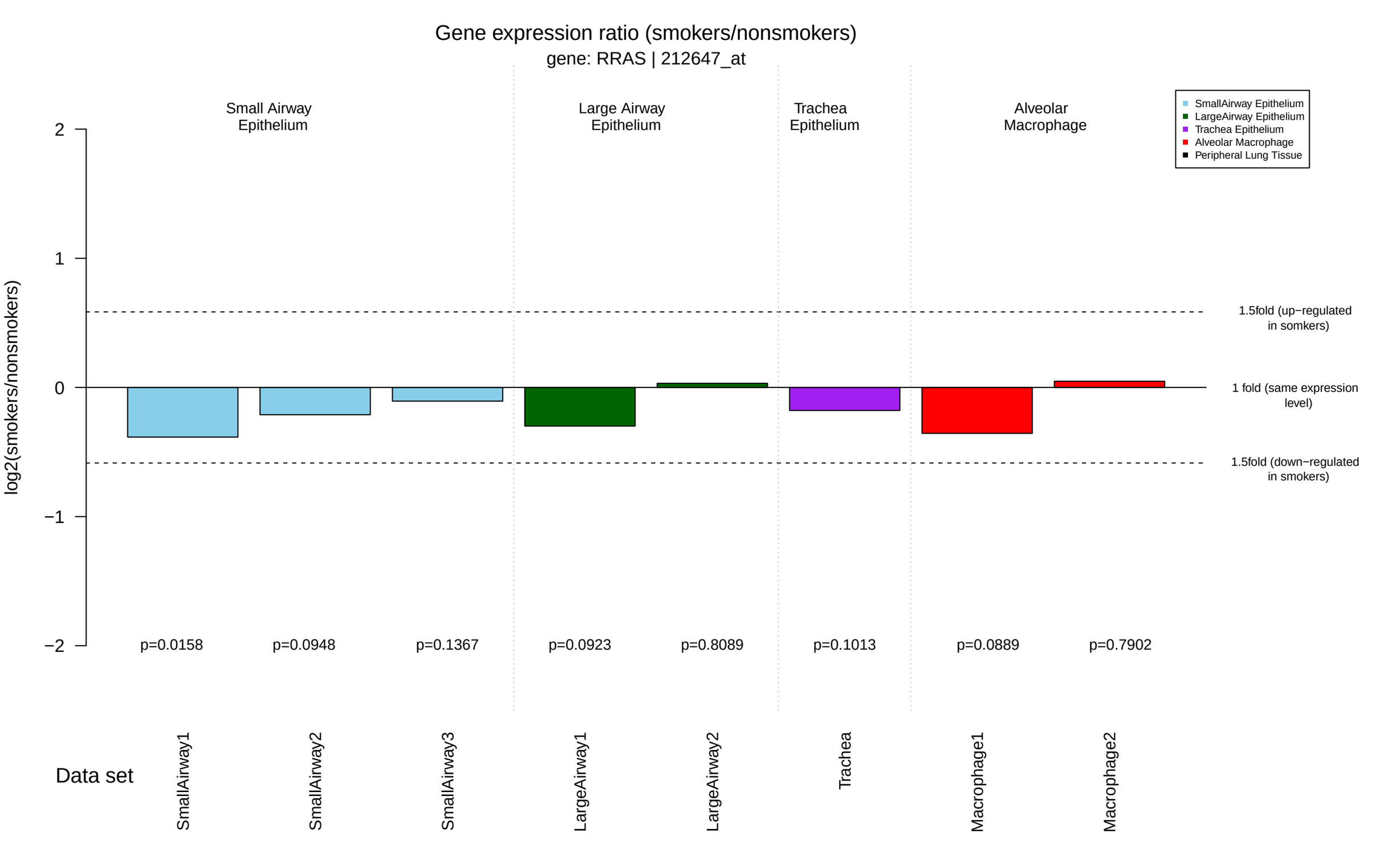
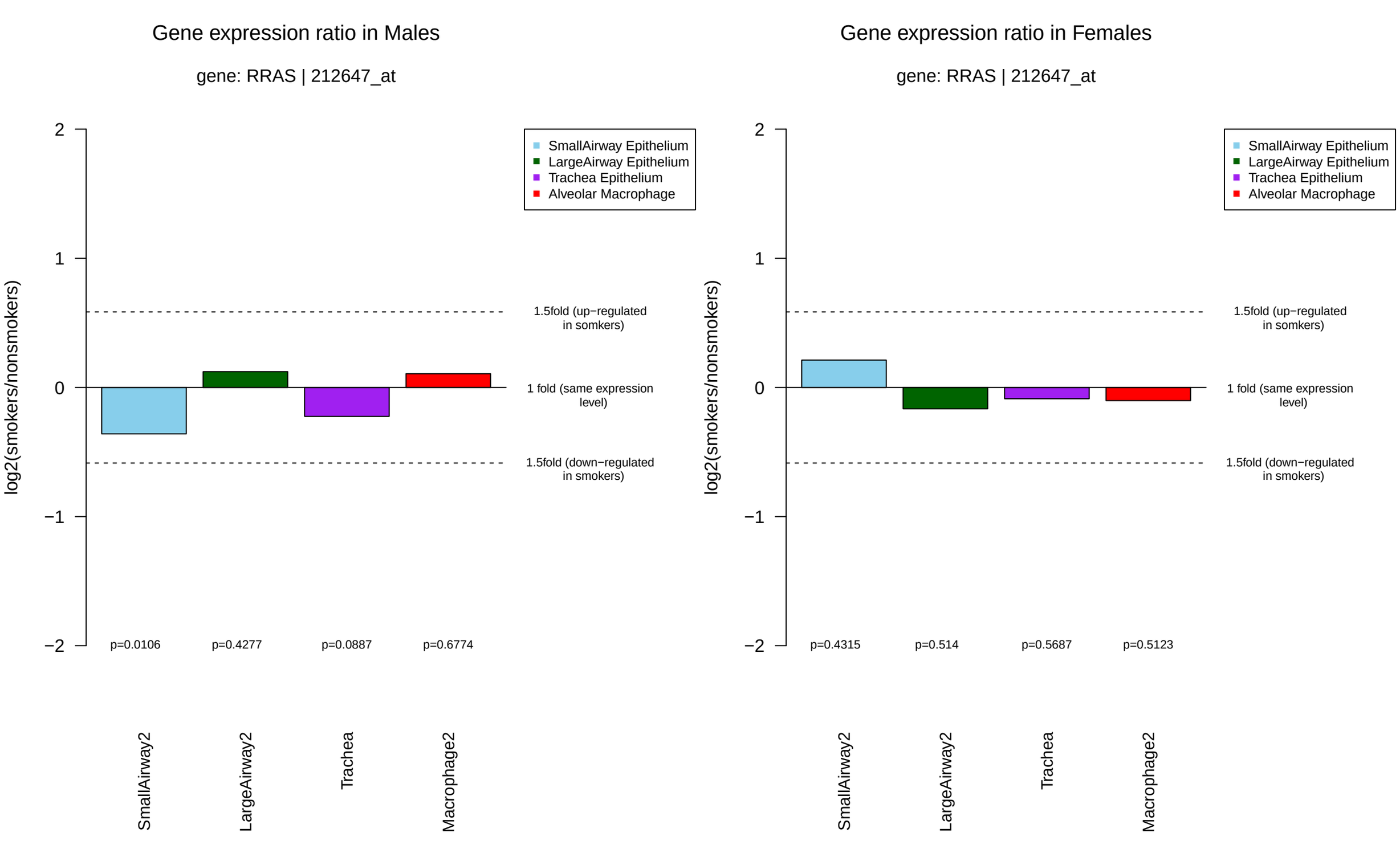
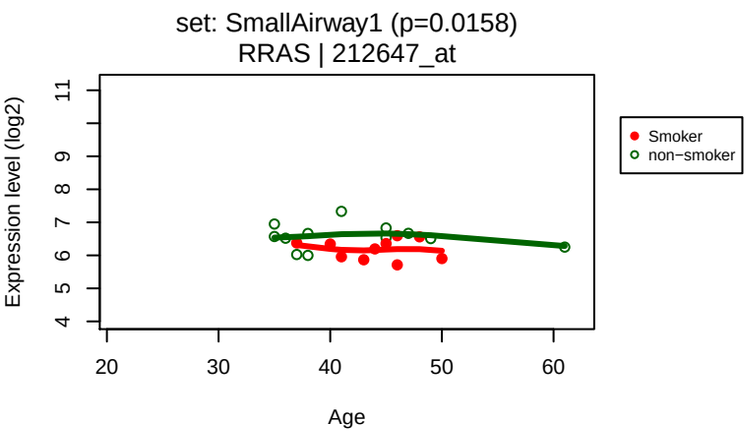
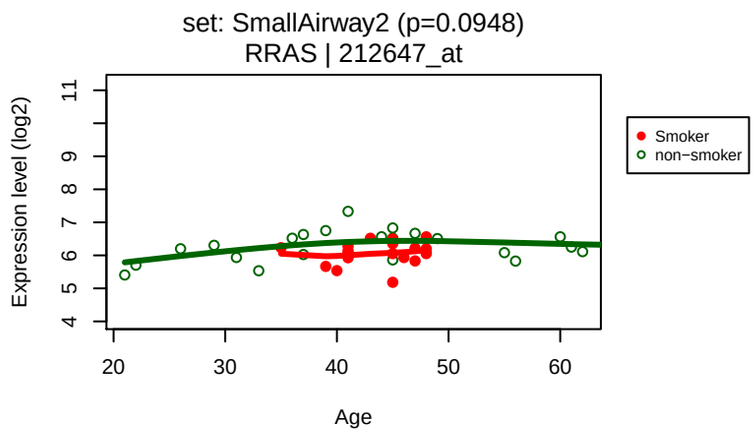
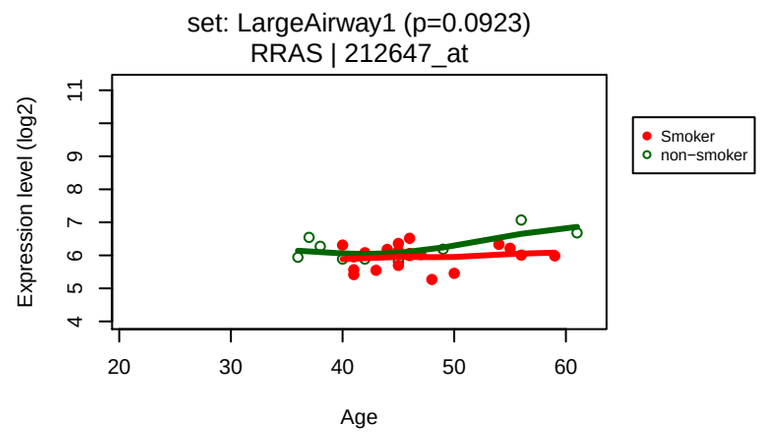

Gene expression of smokers and nonsmokers
gene: RRAS | 212647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0158
p=0.0948
p=0.1367
p=0.0923
p=0.8089
p=0.1013
p=0.0889
p=0.7902
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

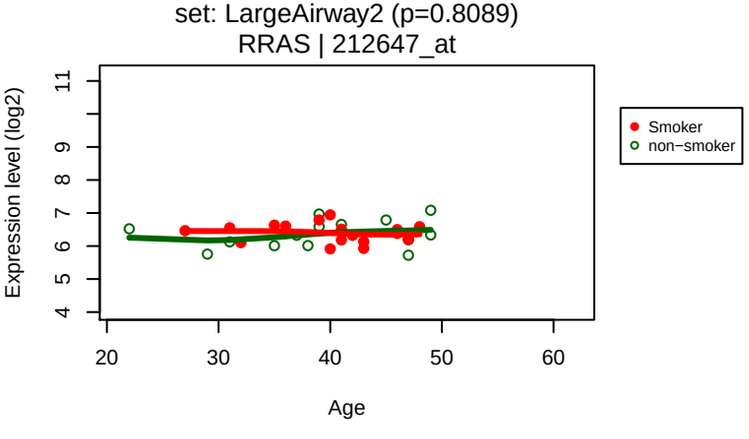

Gene expression ratio (smokers/nonsmokers)
gene: RRAS | 212647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0158
p=0.0948
p=0.1367
p=0.0923
p=0.8089
p=0.1013
p=0.0889
p=0.7902
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

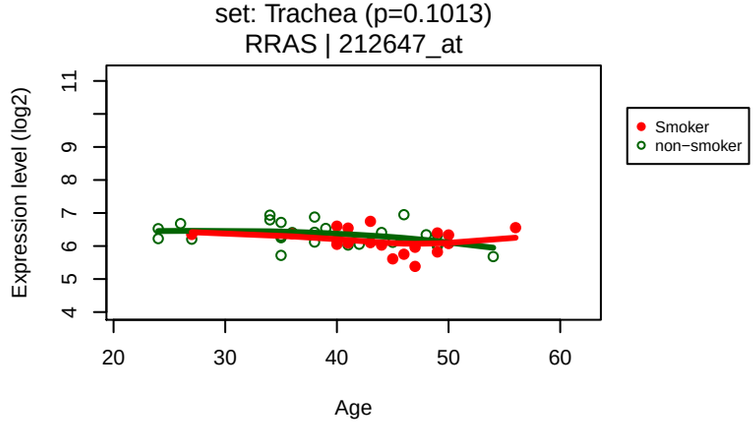

Gene expression ratio in Males
gene: RRAS | 212647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0106
p=0.4277
p=0.0887
p=0.6774
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRAS | 212647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4315
p=0.514
p=0.5687
p=0.5123
SmallAirway2
LargeAirway2
Trachea
Macrophage2

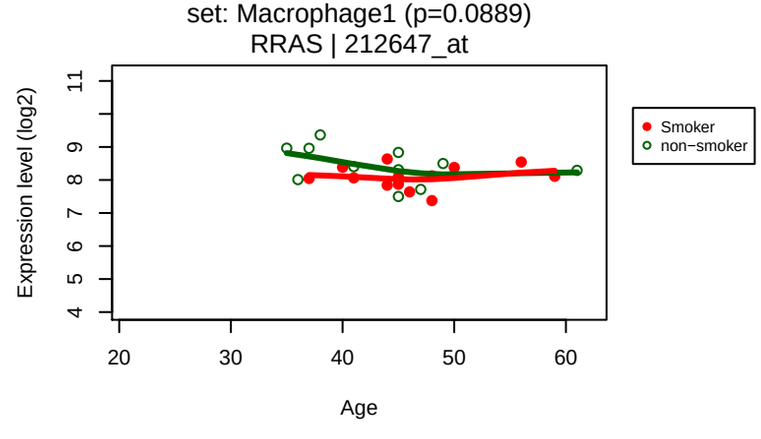

set: SmallAirway1 (p=0.0158)
RRAS | 212647_at
Expression level (log2)
Age
Smoker
non−smoker

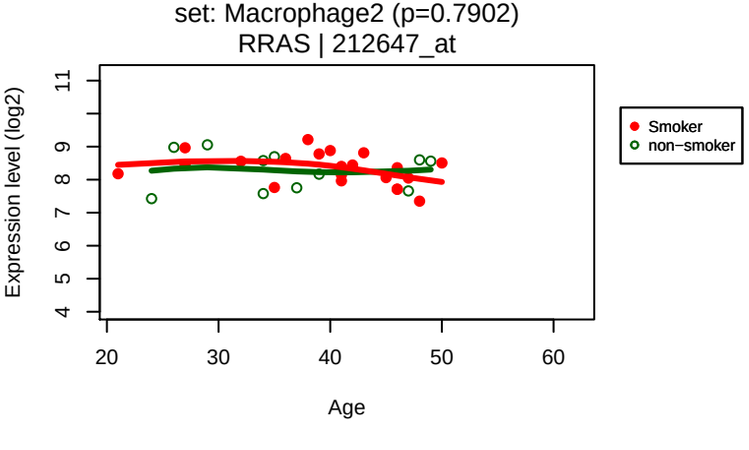

set: SmallAirway2 (p=0.0948)
RRAS | 212647_at
Expression level (log2)
Age
Smoker
non−smoker

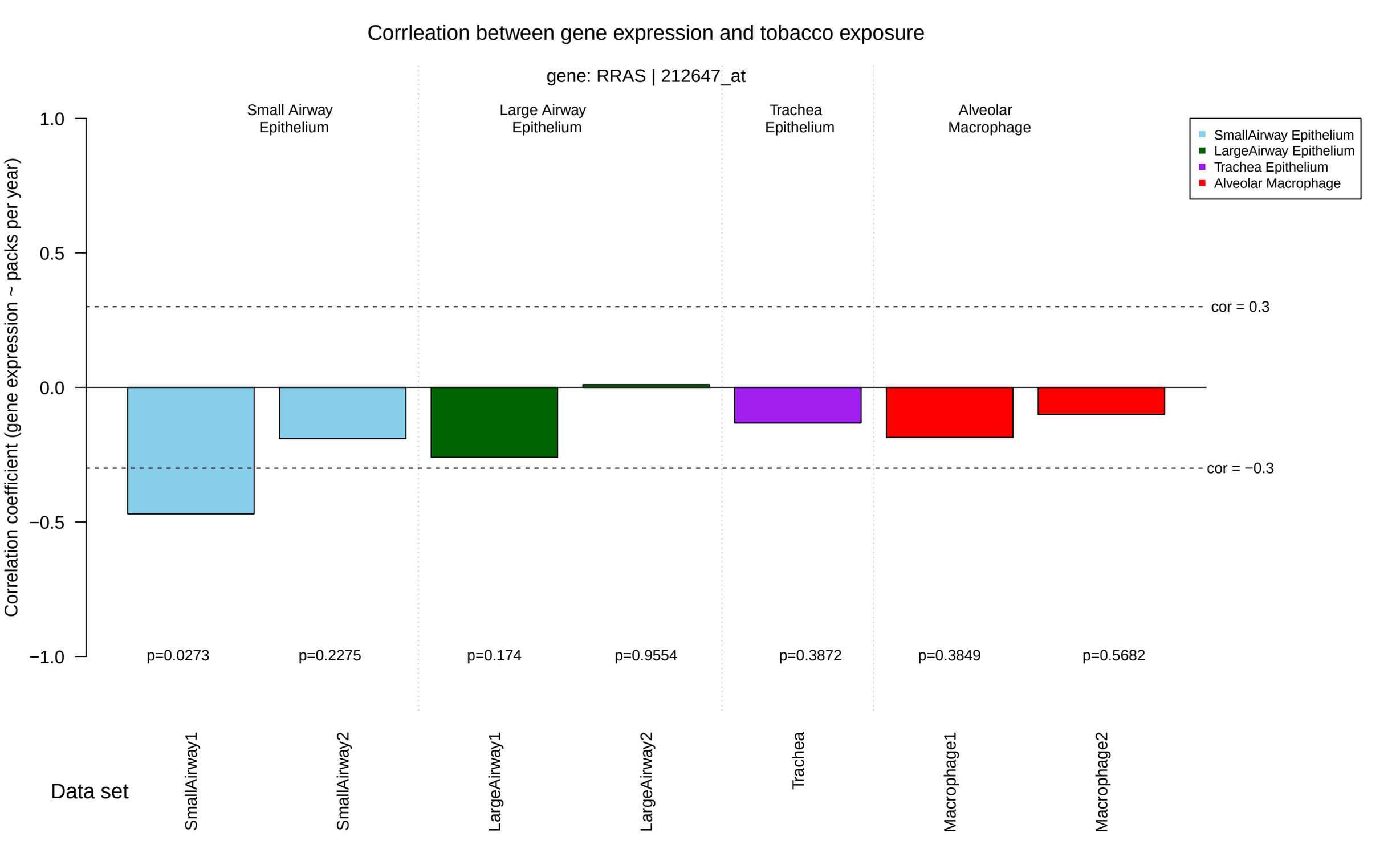

set: LargeAirway1 (p=0.0923)
RRAS | 212647_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8089)
RRAS | 212647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1013)
RRAS | 212647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0889)
RRAS | 212647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7902)
RRAS | 212647_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRAS | 212647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0273
p=0.2275
p=0.174
p=0.9554
p=0.3872
p=0.3849
p=0.5682
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2